

AGA KHAN UNIVERSITY EXAMINATION BOARD
HIGHER SECONDARY SCHOOL CERTIFICATE
CLASS XI EXAMINATION 2009

Physics Paper II

Time allowed: 2 hours 20 minutes Marks 55

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign that it is correct.

I agree that this is my name and school.
Candidate's signature

2. RUBRIC. There are THIRTEEN questions. Answer ALL THIRTEEN questions. Questions 11, 12 & 13 each offers TWO choices. Attempt any ONE choice from each.
3. When answering the questions:

Read each question carefully.
You may use a black pencil for diagrams. DO NOT use coloured pencils.
DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
Complete your answer in the allocated space only. DO NOT write outside the answer box.
4. The marks for the questions are shown in brackets ().

Q.1. (Total 5 Marks)

a. Prove that $2aS = V_f^2 - V_i^2$ is dimensionally correct. (3 Marks)

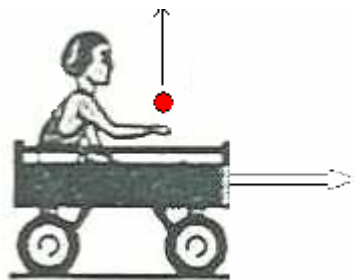
b. Write any TWO differences between precision and accuracy. (2 Marks)

Q.2. (Total 5 Marks)

a. Find $\vec{A} \times \vec{B}$ if $\vec{A} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{B} = 3\hat{i} + 4\hat{k}$. (3 Marks)

- b. Name TWO different conditions that could make $\vec{A}_1 \times \vec{A}_2 = 0$. (2 Marks)

- Q.3. (Total 4 Marks)

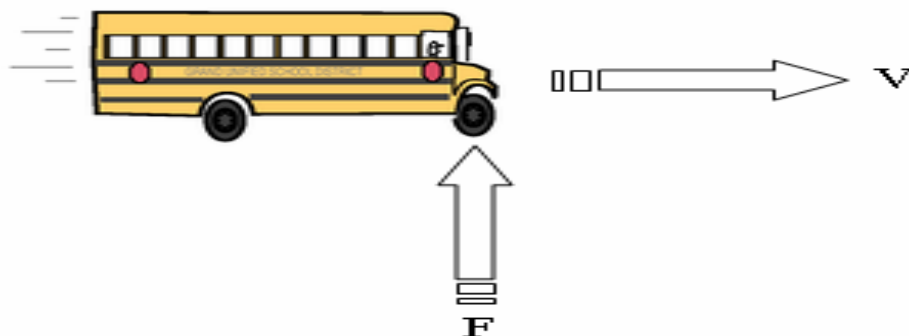


In the above figure a boy is sitting in a cart, moving to the right side at constant velocity. When he throws a ball straight in the upward direction and continues to move in the forward direction, will the ball land in the cart or not? Justify your answer.

Q.4. (Total 3 Marks)

Two cars having same mass of 150kg. The first car changes its velocity from 5 to 10m/s and the second car changes its velocity from 10 to 15m/s. Find out which car has more kinetic energy.

Q.5. (Total 4 Marks)



A bus is moving on a straight path with constant velocity, and additional force perpendicular to its path is applied as shown below

a. Describe what will happen to its motion and what is the name of this perpendicular force? (2 Marks)

- b. On the curved path of the super highway between Karachi and Hyderabad, why are the outer edges of the highway built higher than the inner edges. (2 Marks)

Q.6. (Total 4 Marks)

- a. Write any TWO differences between stream line and turbulent flow. (2 Marks)

- b. In the given diagram water is flowing from a pipe whose diameter is of 2m and the speed of water at this end is 22m/s. If the speed increases to 58m/s then what is the diameter of this portion of pipe? (2 Marks)

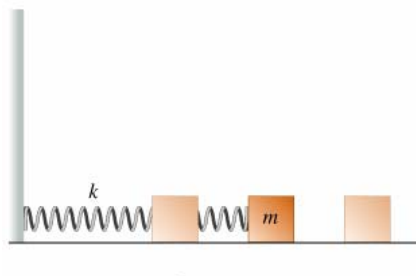


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Q.7.

(Total 5 Marks)

A mass spring system is lying on a smooth horizontal surface as shown below.



The system is executing simple harmonic motion.

- a. Acceleration (a) and displacement (x) are related by the relation, $a \propto -x$

Explain the significance of the negative sign in the relation.

(1 Marks)

- b. A block of weight 19.6N extends the vertical spring by 0.04m from its unstretched position. The block is then replaced by a 0.5kg mass and allowed to come to rest in the new position of the spring. If the spring is now further stretched and then released, calculate the frequency of vibration.

(4 Marks)

Q.8.

(Total 3 Marks)

Define the following terms:

- a. Interference of sound.
- b. Beats.
- c. Stationary waves.

Q.9.

(Total 4 Marks)

Describe the working of Michelson's interferometer with the help of a labelled diagram.

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Q.10.

(Total 3 Marks)

State any THREE postulates of kinetic molecular theory.

Q.11.

(Total 5 Marks)

EITHER

- a. Define and derive an expression for escape velocity and also calculate its value on the surface of earth.

OR

- b. Derive an expression for absolute potential energy.

[illegible]

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EITHER

- OR**

[illegible]

Q.13.

(Total 5 Marks)

EITHER

- a. State the 1st law of thermodynamics and describe its application in
- Isothermal compression.
 - Adiabatic expansion.

OR

- b. Prove that $C_p - C_v = R$

END OF PAPER

Please use this page for rough work